VIRTUAL FACTORY VALIDATION SOFTWARE

IntLearn Kinesis

Validate in a virtual factory before you install

Stack robots, conveyors and cameras like blocks to build a virtual cell (a virtual work area)
Train on defect images made virtually · 8 steps from the virtual check to the real line



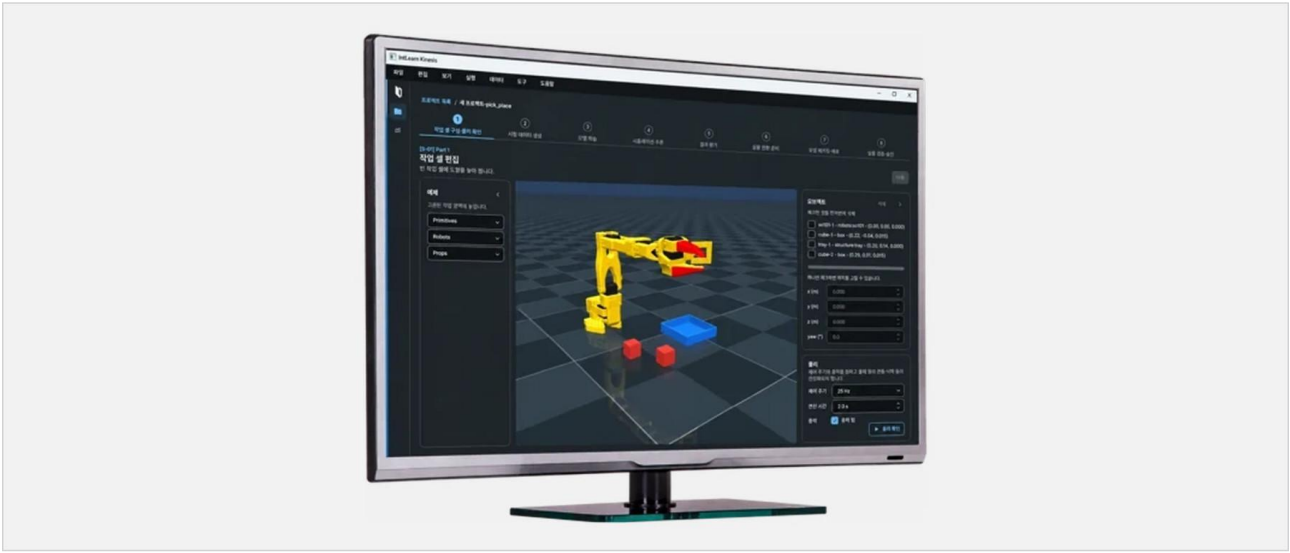
A virtual cell built with IntLearn Kinesis — robot · conveyor · four cameras · staged image

02 — OVERVIEW

Build the cell virtually, check it before the real line

Build robots, conveyors and cameras as a virtual cell, train on synthetic data instead of waiting for defects, and check performance before going on site. IntLearn Kinesis is a new product launched in 2026.

FIG. 01 SOFTWARE



IntLearn Kinesis software screen · example virtual cell (Korean UI)

8 steps	4 views	Patent	2026
Virtual to real line	**Virtual cameras**	**Sim-to-Real**	**New release**
From cell to approval	What each camera sees	Virtual validation · filed	New product, 2026

Key features

One by one on the next pages

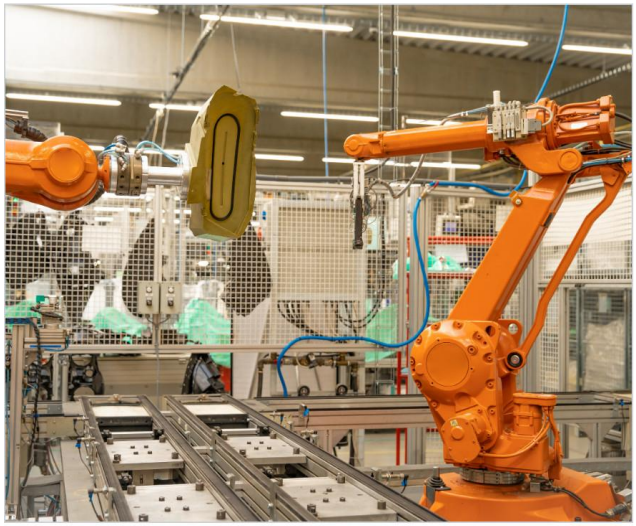
01	**Virtual cell**	Robot, conveyor, four cameras, placed like blocks
02	**Synthetic data**	Make defects instead of waiting for them
03	**Training · simulated inference**	Run the trained AI in the virtual cell before going on site
04	**Real-world transfer · approval**	Take what was validated virtually to the real line
05	**Patent application**	Sim-to-Real virtual validation: check virtually, then apply

03 — WHY VIRTUAL FIRST

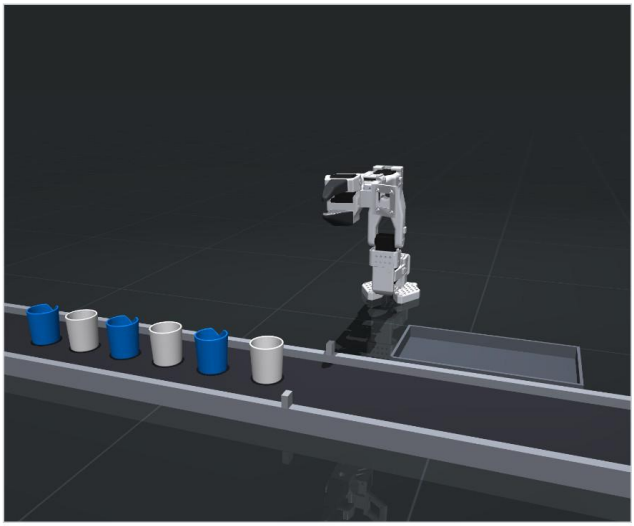
What if you could check it before it reaches the line?

Often you only learn whether inspection equipment works after it is installed on the line, and the defect images needed for training come only by waiting for real defects. IntLearn Kinesis changes that order: stack robots, conveyors and cameras like blocks to build a virtual cell, check everything there first, then apply it on the real line.

FIG. 02 REAL LINE · VIRTUAL CELL



On site · a real line, after installation



Virtual cell · placed before anything is installed

01

No waiting for defects

Make defect images in the virtual cell and train the AI on them.

02

Check before installing

Run the trained AI in the virtual cell and see how it performs first.

03

Virtual to the real line

Take it to the real line, check it again there and approve it.

Validate in a virtual factory before you install.

It starts with placing robots, conveyors and cameras in a virtual cell.

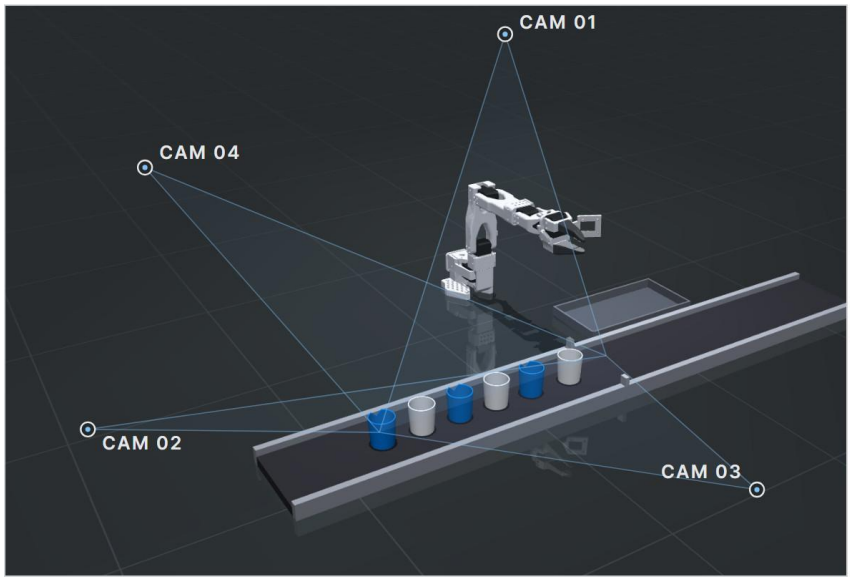
04 — FEATURES · 01 — 02

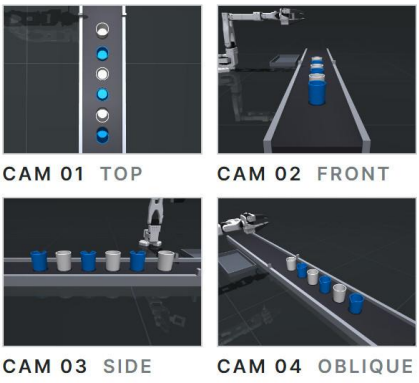
Stack the cell like blocks, then make the training images

01 Virtual cell Robot · conveyor · four camera views

Place the robot, conveyor, parts and cameras like blocks, and see what each camera sees before installing.

FIG. 03 VIRTUAL CELL · 4 VIEWS





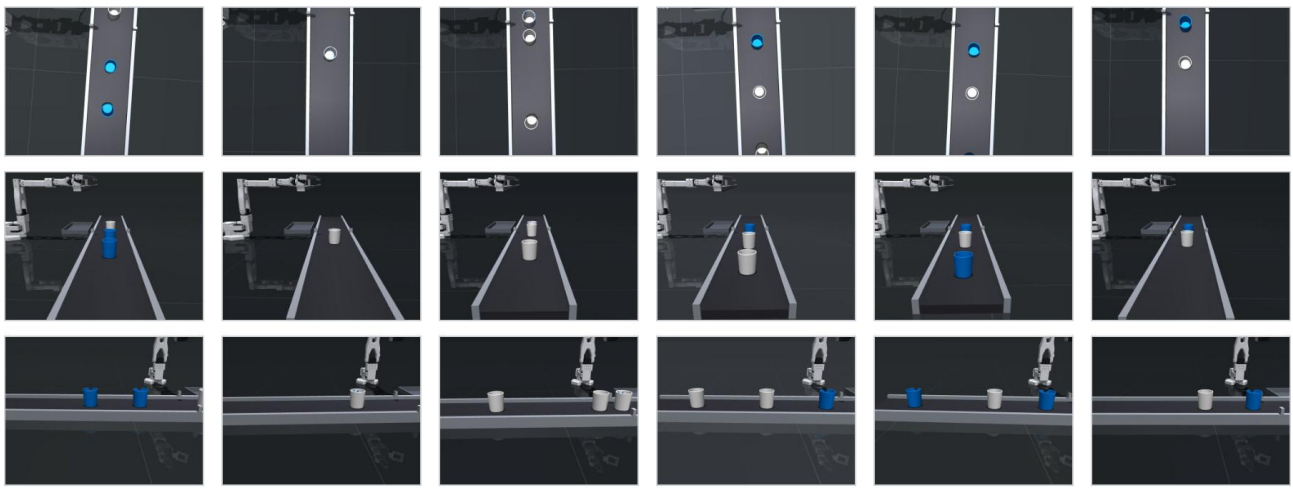
What each of the four cameras in the cell on the left sees. Of the 6 cups, the 3 in the accent color are chipped (defective).

IntLearn Kinesis example 'Conveyor stop and sort' cell · what the four cameras see

02 Synthetic data Make defects instead of waiting for them

Make training images (synthetic data) in the virtual cell, so there is no waiting for real defects.

FIG. 04 SYNTHETIC DATA · 6 LAYOUTS × 3 CAMERAS



Synthetic data — one virtual cell, with cup position, count, defects, lighting and camera varied · blue cup = chipped rim

05 — FEATURES · 03 — 05

In the virtual cell, run it all the way through

03 **Training · simulated inference** Check performance before going on site

Train the AI on the images you made, then run the trained AI inside the virtual cell (simulated inference) to check performance before going on site.

FIG. 05 SIMULATION · ONE EPISODE



01 ENTERS 02 MOVES 03 STOPS

04 ARM DOWN 05 GRIPS 06 LIFTS

One run in the virtual cell, start to finish — the robot arm picks up the object that came in on the conveyor
A pre-scripted motion shown as a demonstration, not the result of a trained AI

04

Real-world transfer · approval

Take what you checked virtually to the real line, check it again on the real thing and approve it.

05

Patent application

Sim-to-Real virtual validation (check virtually, then apply on the real line)

Check performance virtually, before the line.

Check it virtually first, then move it to the real line.

06 — EVIDENCE

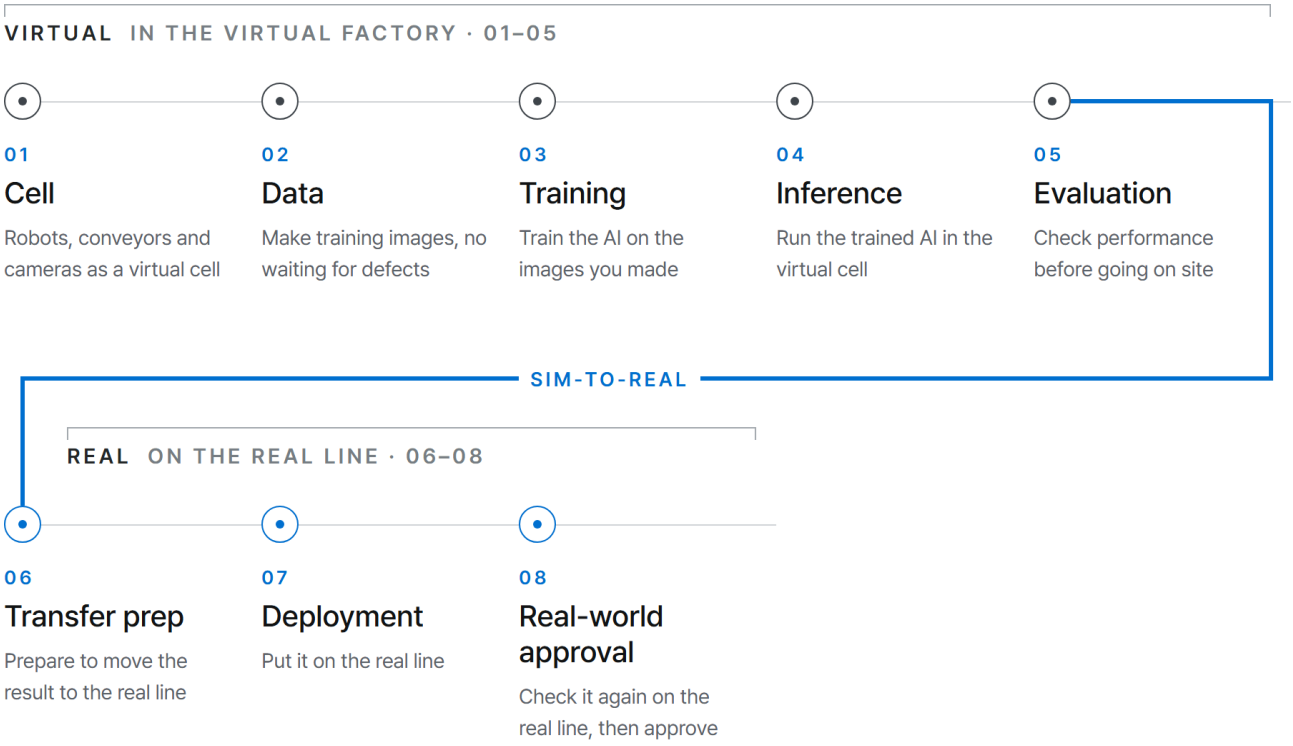
From cell to real-world approval, in eight steps

8 steps

Sim-to-Real pipeline

The order: check virtually, then apply on the real line

Cell → data → training → inference → evaluation → transfer prep → deployment → real-world approval



IntLearn Kinesis Sim-to-Real pipeline, 8 steps

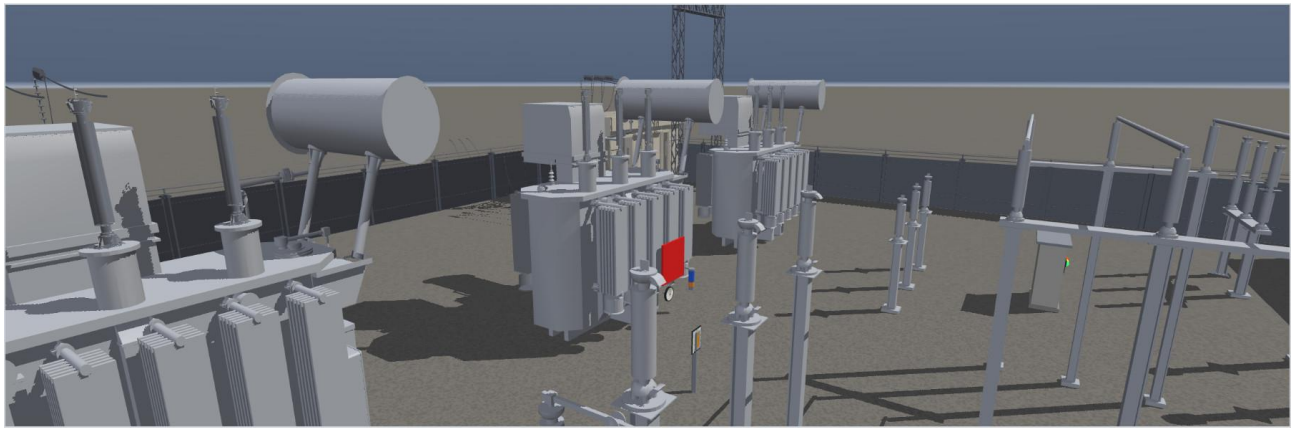
Virtual first, the real line next.

The cell, training images and trained AI built in the virtual factory lead to checks and approval on the real line.

07 — VIDEO · PLATFORM

Watch it, then connect it

FIG. 07 VIRTUAL SUBSTATION · PATROL



IntLearn Kinesis example 'Outdoor substation patrol': a robot patrols a virtual substation and checks equipment

Equipment 3D models: kevinhalim9 · Gazebo Fuel · CC BY 4.0

Product videos

Point your phone camera at a QR code

Main · 70 s · simulation

What Does a Robot Dog Inspect at a Substation?

Series · 40 s

Build a Virtual Substation for Robot Dog Patrols

Series · 35 s

Three Robot Dog Models, One Virtual Workspace

YouTube @intelliz

Watch more on YouTube

Connected products on the platform



IntCloud Optic

An AI optical inspection module that captures defects so they can be seen



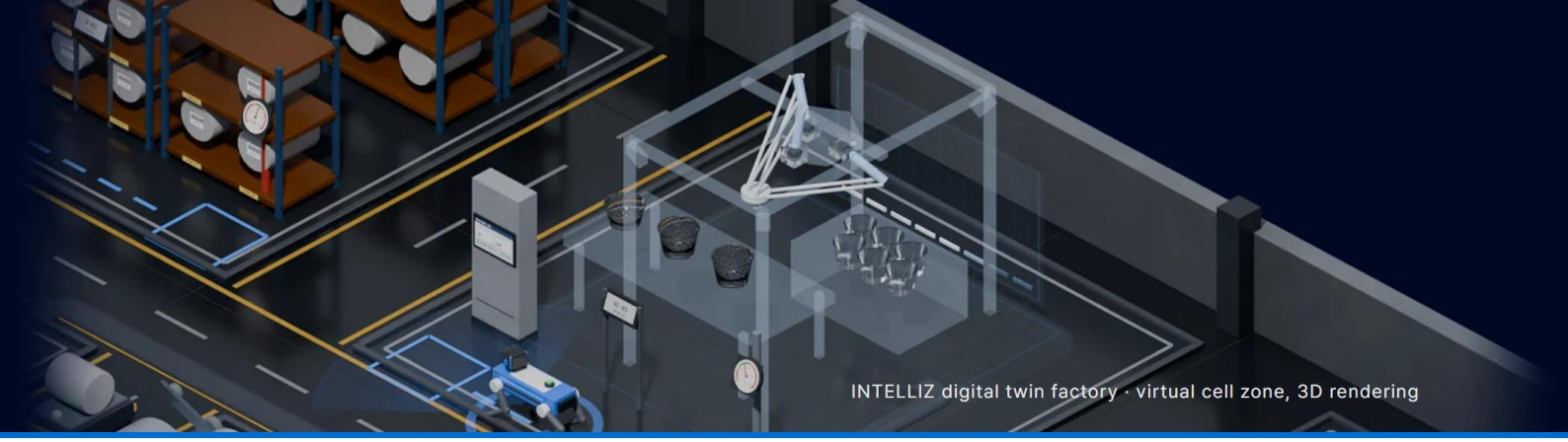
IntLearn Pro

Deep learning inspection software that finds defects like an expert, with 12 engines and modules, the most in Korea, and LLM integration



IntWise New in 2026

Operate by speaking, and the know-how becomes a company asset



INTELLIZ digital twin factory · virtual cell zone, 3D rendering

08 — CONTACT · NEXT STEPS

Considering a similar inspection?

Including the following in your inquiry helps us prepare our answer.

01	What you inspect: material, shape and size	02	The defects to find and your pass/fail criteria
03	Line speed and how products are conveyed	04	Space available for cameras and lighting
05	Where results need to go (PLC, reject unit, etc.)		

Next steps

Addresses on intelliz.net

Guide	AI Visual Inspection vs Rule-Based Machine Vision	intelliz.net/en/guides/ai-visual-inspection-basics
Optics Lab	Can Line Scan or Area Scan Inspect It?	intelliz.net/en/solutions/optics-lab
Getting ready	AI Inspection Adoption Checklist	intelliz.net/en/tools/inspection-checklist
FAQ	Can performance be checked before installation?	intelliz.net/en/company/faq#before-install

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Japan distributor BIT DEVICE Co., Ltd., 6-3-24-L430 Nishinakajima, Yodogawa-ku, Osaka, Japan

Product page

Try it

INTELLIZ

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Virtual cell images are staged for this brochure with the IntLearn Kinesis engine; the photo is for illustration.
SO-101 robot model: MuJoCo Menagerie (Apache-2.0)
Substation equipment models: kevinhalim9 / Gazebo Fuel (CC BY 4.0)